# Lecture 2: The Simplex Method & Diameter of Polyhedra

## Algorithms & Geometry of Linear Programming

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# From geometry to algorithm

Lecture 1: a feasible, bounded LP is solved at a *vertex*, and local optimality implies global optimality.

Today: the simplex method [Dantzig '47] — walk from vertex to vertex along improving edges.

- **in practice**: number of pivots typically *linear* in the problem size [Shamir '87]
- **in theory**: exponential worst case: the Klee–Minty cube [Klee–Minty '72]

Second theme: how short can vertex-to-vertex paths be? — the diameter of polyhedra.

# Today

The Landscape of LP Algorithms

The Simplex Method

Pivot Rules

The Diameter of Polyhedra

# Outline

The Landscape of LP Algorithms

The Simplex Method

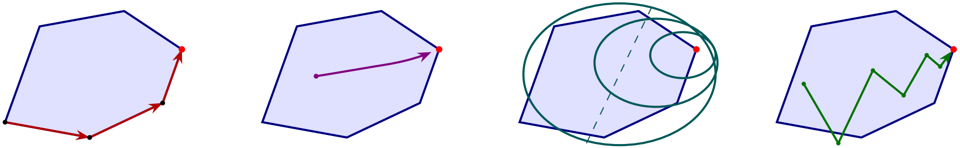
Pivot Rules

The Diameter of Polyhedra

# Four families of LP algorithms

Two equivalent formulations; data $(\mathbf{A}, b, c)$, $\mathbf{A} \in \mathbb{R}^{m \times n}$; $L :=$ bit length:

$\min\ c^{\mathsf{T}} x$ s.t. $\mathbf{A}x = b,\ x \geq \mathbf{0}$ (standard) $\qquad \max\ c^{\mathsf{T}} x$ s.t. $\mathbf{A}x \leq b$ (geometric)



simplex [Dantzig '47] — interior point [Karmarkar '84] — ellipsoid [Khachiyan '79] — first-order

Each iteration is a fixed amount of linear algebra $\Rightarrow$ compare iteration counts.

# Exact versus approximate solvers

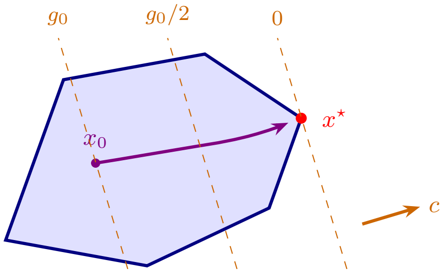
exact: terminate with an optimal vertex + certificate *(simplex)*

*How many iterations to reach the optimum?*

approximate: drive down the gap $g(x) := c^\mathsf{T} x - \mathrm{OPT}$ *(interior point, ellipsoid, first-order)*

*How many iterations to halve the error?*

each method reduces a computable proxy: barrier parameter $\mu$ / ellipsoid volume / KKT residual




halving the gap = advancing one dashed line

# From weakly to strongly polynomial

- over the rationals, approximate $\Rightarrow$ exact: gap below $2^{-\Theta(L)}$ $\Rightarrow$ round to the optimal vertex
- halving the gap every $T$ iterations solves the LP in $O(T \cdot L)$ iterations: weakly polynomial

Definition (Strongly polynomial algorithm [Megiddo '83])

An exact LP algorithm is *strongly polynomial* if (i) it performs $\mathrm{poly}(m, n)$ basic arithmetic operations, and (ii) intermediate numbers stay of bit length $\mathrm{poly}(L)$.

- existence of one for LP: major open problem, on Smale's list for the 21st century [Smale '98]
- a $\mathrm{poly}(m, n)$ pivot bound for *some* pivot rule would resolve it

# Strongly polynomial algorithms: what is known

- maximum flow [Edmonds–Karp '72], minimum-cost flow [Tardos '85]
- $\mathbf{A}$ with polynomially bounded integer entries [Tardos '86] — running time independent of $b$ and $c$
- interior point with iteration count depending only on $\mathbf{A}$ [Vavasis–Ye '96], via a condition measure $\approx$ the *circuit imbalance* $\kappa_{\mathbf{A}}$ of Lecture 3 [DHNV '20]
- $\leq 2$ nonzeros per column of $\mathbf{A}$ [DKNOV '24]
- a limit: no central-path-following interior point method is strongly polynomial [ABGJ '18] — Lecture 4

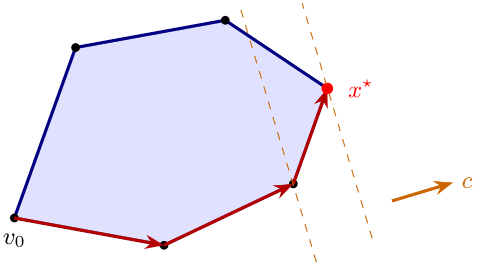
Today: the exact side of the landscape — the simplex method.

# Outline

# The vertex-edge graph, and the walk

Throughout: $P = \{x : \mathbf{A}x \leq b\}$, $\mathrm{rank}(\mathbf{A}) = n$, LP $\max\{c^\mathsf{T}x : x \in P\}$.

*vertex-edge graph* $G(P)$: nodes = vertices, edges = bounded 1-dim. faces




## Abstract simplex method

While an incident edge *improves* the objective, move along it (ray $\Rightarrow$ *unbounded*). Output the final vertex.

choice of improving edge = the pivot rule

To justify: stuck $\Rightarrow$ optimal, and the walk terminates.

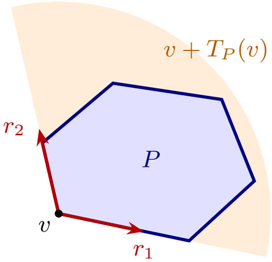
# The local object: the tangent cone

**Definition (Tangent cone)**

For $x \in P$ with tight constraints $I(x) = \{i : a_i^\mathsf{T} x = b_i\}$:

$$T_P(x) := \{w : x + \varepsilon w \in P \text{ for some } \varepsilon > 0\} = \{w : \mathbf{A}_{I(x)} w \leq \mathbf{0}\}.$$

seen from $x$, all of $P$ lies in the cone:
$P \subseteq x + T_P(x)$




**Lemma (Edges generate the tangent cone)**

*At a vertex $v$: $T_P(v)$ is pointed and*

$$T_P(v) = \mathrm{cone}\left(e - v \ : \ e \text{ edge of } P \text{ at } v\right),$$

*with extreme rays $=$ the incident edge directions.*

# Correctness of the simplex method

**Theorem**

*Let $v \in V(P)$. Exactly one of the following holds:*

(a) $c^\mathsf{T} r \leq 0$ *for every incident edge direction* $r$*: then* $v$ *is optimal;*

(b) *some incident edge direction has* $c^\mathsf{T} r > 0$*: then moving along it reaches a strictly better vertex, or certifies that the LP is unbounded.*

**Proof.** (a) For $y \in P$: $\quad y - v \in T_P(v) = \mathrm{cone}(\text{edge directions})$, say $y - v = \sum_r \mu_r r$, $\mu \geq \mathbf{0}$, so

$$c^\mathsf{T} y - c^\mathsf{T} v = \sum_r \mu_r \, c^\mathsf{T} r \leq 0.$$

(b) The edge ends at a neighbor of strictly larger value, or is a ray $r \in \mathrm{rec}(P)$ with $c^\mathsf{T} r > 0$ — unbounded (Lecture 1). $\square$

# Termination — and degeneracy

### Corollary (termination)

If every pivot strictly increases the objective, no vertex is revisited: at most $|V(P)| \leq \binom{m}{n}$ pivots, *for any pivot rule*.

- strict increase is guaranteed when no vertex has $> n$ tight constraints
- degeneracy: $> n$ tight constraints — a pivot may exchange tight constraints *without moving*; the pivot rule can cycle [Bland '77: anti-cycling]
- cure (exercise): perturb $b \mapsto b + (\varepsilon, \varepsilon^2, \ldots, \varepsilon^m)^\mathsf{T}$ — no vertex degenerate for small $\varepsilon > 0$, and an optimal basis of the perturbed LP is optimal for the original

We assume non-degeneracy today. The real question: how many pivots?

# A byproduct: the vertex-edge graph is connected

**Corollary**

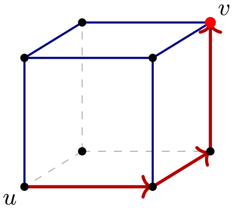
If $P$ is pointed and non-empty, then $G(P)$ is connected.

**Proof.** To reach a target vertex $v$ from any $u$: build an objective maximized *uniquely* at $v$,

$$c := \sum_{i \in I(v)} a_i \implies c^\mathsf{T} y \le \sum_{i \in I(v)} b_i = c^\mathsf{T} v \quad \forall y \in P,$$

equality iff $y = v \quad (\operatorname{rank} \mathbf{A}_{I(v)} = n)$. Run simplex from $u$: bounded LP $\Rightarrow$ no rays taken; strict increase $\Rightarrow$ terminates — at the unique optimum $v$. $\square$

The walk itself traces the $u$–$v$ path.




1-skeleton of the cube

# The simplex method in standard form

Geometric form is the language of pictures; for *computation*, switch to

$$\min c^{\mathsf{T}} x \text{ s.t. } \mathbf{A}x = b, \quad x \geq \mathbf{0}, \qquad \mathbf{A} \in \mathbb{R}^{m \times n} \text{ of full row rank.}$$

| **geometric form** $\{x : \mathbf{A}x \leq b\}$ | **standard form** $\{x : \mathbf{A}x = b,\ x \geq \mathbf{0}\}$ |
|---|---|
| tight constraint $a_i^{\mathsf{T}} x = b_i$ | zero coordinate $x_j = 0$ |
| vertex $= n$ indep. tight constraints | BFS: $x_N = \mathbf{0}$, $\mathbf{A}_B$ invertible |
| edge $= n - 1$ indep. tight constraints | $x_{[n]\setminus S} = \mathbf{0}$, $\ker(\mathbf{A}_S)$ 1-dim. ($\lvert S\rvert = m+1$) |
| tangent cone $\{w : \mathbf{A}_{I(x)} w \leq \mathbf{0}\}$ | $\{w \in \ker(\mathbf{A}) : w_j \geq 0 \text{ if } x_j = 0\}$ |
| non-deg.: no vertex with $> n$ tight | non-deg.: $x_B > \mathbf{0}$ at every BFS |

via *slack coordinates* $s := b - \mathbf{A}x$: constraint $i$ tight $\iff$ $s_i = 0$

# Edge directions and reduced costs: algebra $=$ geometry

Basis $B \subseteq [n]$, $|B| = m$, $\mathbf{A}_B$ invertible; BFS $x$: $x_B = \mathbf{A}_B^{-1} b \geq \mathbf{0}$, $x_N = \mathbf{0}$.
Non-degenerate ($x_B > \mathbf{0}$): tight constraints $= \{\mathbf{A}x = b\} \cup \{x_j = 0 : j \in N\}$.

Incident edge $=$ relax *one* bound $x_j = 0$, keep the rest tight:

$$\mathbf{A}w^{(j)} = \mathbf{0}, \quad w^{(j)}_{N\setminus\{j\}} = \mathbf{0}, \quad w^{(j)}_j = 1 \qquad \Longrightarrow \qquad w^{(j)}_B = -\mathbf{A}_B^{-1}\mathbf{A}e_j$$

objective slope along the edge, *per unit of* $x_j$:

$$c^{\mathsf{T}} w^{(j)} \;=\; c_j - c_B^{\mathsf{T}}\mathbf{A}_B^{-1}\mathbf{A}e_j \;=:\; \bar{c}_j \qquad (\text{reduced cost};\ \bar{c}_B = \mathbf{0})$$

$n - m$ directions, one per non-basic $j$ $=$ extreme rays of $\{w \in \ker(\mathbf{A}) : w_N \geq \mathbf{0}\}$.
Optimality test, fully explicit: $\bar{c} \geq \mathbf{0} \iff x$ optimal.

# One iteration, in full

Simplex method, standard form, non-degenerate (one iteration)

Given: basis $B$ with BFS $x$, $\quad x_B = \mathbf{A}_B^{-1} b > \mathbf{0}$.

1. reduced costs $\bar{c}^{\mathsf{T}} = c^{\mathsf{T}} - c_B^{\mathsf{T}} \mathbf{A}_B^{-1} \mathbf{A}$; $\quad$ if $\bar{c} \geq \mathbf{0}$: $\quad x$ optimal — stop
2. *entering* index: choose $j$ with $\bar{c}_j < 0$ $\quad$ (the pivot rule)
3. edge direction: $w_j = 1, \quad w_B = -\mathbf{A}_B^{-1} \mathbf{A} e_j,$ $\quad$ zero elsewhere
4. if $w_B \geq \mathbf{0}$: $\quad$ *unbounded* along the ray $x + t\, w$ — stop
5. ratio test: $t^\star = \min_{i \in B:\, w_i < 0} x_i / (-w_i)$, attained at *leaving* index $i^\star$
6. update $\quad x \leftarrow x + t^\star w, \quad B \leftarrow B \cup \{j\} \setminus \{i^\star\}$

cost per pivot: one linear system solve; $\quad$ non-degeneracy $\Rightarrow t^\star > 0 \Rightarrow$ termination

# Initialization: the two-phase scheme

The method needs a starting BFS — but finding *any* feasible point is a priori as hard as LP itself.

Resolution: feasibility is itself an LP *with an obvious starting vertex*.

Phase 0 (after flipping signs of rows so that $b \geq \mathbf{0}$)

$$\min \mathbf{1}^\mathsf{T} z \text{ s.t. } \mathbf{A}x + z = b, \quad x, z \geq \mathbf{0}, \qquad \text{starting BFS: } (x, z) = (\mathbf{0}, b)$$

- optimal value $0 \iff$ the original LP is feasible
- an optimal BFS with $z = \mathbf{0}$ starts *Phase 1*: simplex on the real objective

# Where we are

**This hour:**

- simplex method: walk along improving edges — correct for every pivot rule (tangent cone), terminating under non-degeneracy, $G(P)$ connected
- fully explicit in standard form: edges $=$ kernel directions, slopes $=$ reduced costs, one linear system per pivot; initialization via two phases

**Wide open so far:** how many pivots?

- does some clever choice of improving edge always take few steps?
- are there always *short* vertex-to-vertex paths at all?

This afternoon: pivot rules and the Klee–Minty cube; then the diameter of polyhedra.

# Outline

# Classical pivot rules

A pivot rule decides which improving edge to take.

- **Dantzig's rule**: most negative reduced cost $\bar{c}_j$ — improvement *per unit* of the entering variable
- **greatest improvement**: maximize the actual decrease $t^\star|\bar{c}_j|$ — needs a ratio test per candidate
- **steepest edge**: edge direction of smallest angle with $-c$ — empirically the strongest; its approximation *Devex* is the default in modern solvers

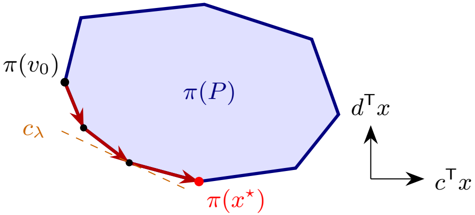
None is known to be efficient in the worst case; for most classical rules, explicit exponential examples exist.

# The shadow vertex rule

Two objectives: $c$ with *known unique minimizer* $v_0$, and a *target* $d$. Slide:

$$c_\lambda := (1-\lambda)\,c + \lambda\,d, \qquad \lambda : 0 \to 1, \qquad \text{maintaining a } c_\lambda\text{-minimizing vertex.}$$

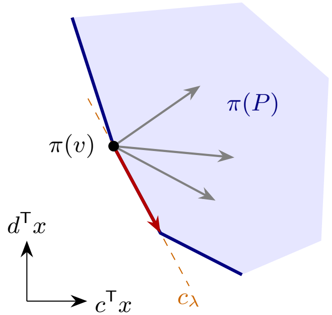
- breakpoints finitely many, consecutive optima adjacent: a simplex path $v_0 \to$ $d$-optimum
- vertex visited $\iff$ it minimizes *some* $c_\lambda$ $\iff$ it projects onto the lower-left boundary of the shadow $\pi(P)$, $\pi(x) = (c^\mathsf{T} x, d^\mathsf{T} x)$
- that boundary = the trade-off curve: best $d$-value for each $c$-value




#pivots $\leq$ #vertices of $\pi(P)$

# Shadow vertex: anchoring, implementation, locality

- **where does $c$ come from?** any $c$ with $c_B = \mathbf{0}$, $c_N > \mathbf{0}$ has unique minimizer $v_0$ — the tight-constraint trick again
- **which edge enters?** among edges progressing in the target ($d^\mathsf{T} w < 0$): minimize the exchange rate $c^\mathsf{T} w / (-\,d^\mathsf{T} w)$ — $c$-cost per unit of $d$-progress = steepest descent in the shadow plane
- **implementation**: $\bar{c}(\lambda)$ is *affine* in $\lambda$ — a ratio test finds the next breakpoint
- **locality**: visited $\iff$ minimizes some $c_\lambda$ — *not on the history of the walk*




edges point into $\pi(P)$: none is steeper
than the boundary edge

Locality is what has enabled the probabilistic analyses of simplex — the trade-off curve returns in Lectures 3 and 4, always traversed from the $c$-optimal end.

# The random facet rule

Best worst-case guarantee known: randomize.  (Geometric form, dimension $n$, $m$ facets.)

## Random facet [Kalai '92]

At vertex $v$: if no improving edge, stop. Else pick a facet $F \ni v$ *uniformly at random*, recursively optimize over $F$ (dimension $n-1$). If $F$'s optimum is optimal for $P$, stop; else take an improving edge out of $F$ and recurse.

- expected pivots $\leq e^{O(\sqrt{n \log m})}$ on *every* LP [Kalai '92; Matoušek–Sharir–Welzl '96] — subexponential
- still the state of the art; improved variant [Hansen–Zwick '15]; nearly matching lower bounds for the rule [Friedmann–Hansen–Zwick '11]

# Exponential examples: the Klee–Minty cube

For a parameter $0 < \varepsilon < 1/2$ [Klee–Minty '72]:

$$\begin{aligned} \max \quad & x_n \\ \text{s.t.} \quad & 0 \le x_1 \le 1, \\ & \varepsilon x_{j-1} \le x_j \le 1 - \varepsilon x_{j-1}, \qquad j = 2, \dots, n \end{aligned}$$

- pair $j$: a floor $\varepsilon x_{j-1} \le x_j$ and a ceiling $x_j \le 1 - \varepsilon x_{j-1}$ — a perturbed cube $[0,1]^n$ with $2n$ facets
- floor and ceiling of a pair are never both tight — each vertex makes *exactly one* of each pair tight
- vertices $v^S \leftrightarrow$ subsets $S \subseteq [n]$ (pairs with tight ceiling), via forward substitution:

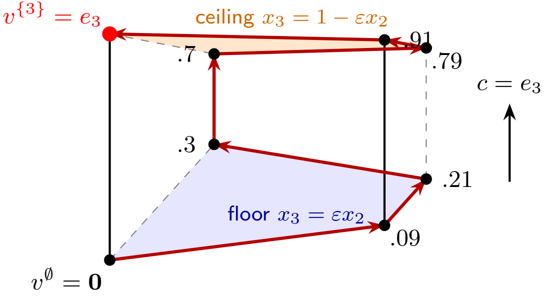
$$v_j^S = \begin{cases} 1 - \varepsilon\, v_{j-1}^S, & j \in S \\ \varepsilon\, v_{j-1}^S, & j \notin S \end{cases} \qquad (v_0^S := 0)$$

# A monotone Hamiltonian path

**Theorem (Klee–Minty '72)**

*For $0 < \varepsilon < 1/2$, the graph of $P_n$ has a Hamiltonian path from $v^{\emptyset} = \mathbf{0}$ to $v^{\{n\}} = e_n$ along which $x_n$ strictly increases.*

every step improves the objective, yet *all* $2^n$ vertices are visited: an adversarial improving pivot rule takes $2^n - 1$ pivots, with only $2n$ facets

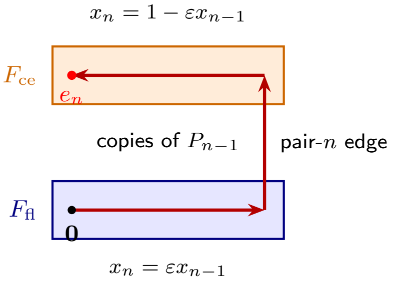



$n = 3$, $\varepsilon = 0.3$, true coordinates; labels = objective values $x_3$

# Proof idea

Induction on $n$:

- every vertex lies on exactly one pair-$n$ facet: floor $F_{\text{fl}} = \{x_n = \varepsilon x_{n-1}\}$ or ceiling $F_{\text{ce}} = \{x_n = 1 - \varepsilon x_{n-1}\}$
- both are *affine copies* of $P_{n-1}$: $y \mapsto (y, \varepsilon y_{n-1})$ and $y \mapsto (y, 1 - \varepsilon y_{n-1})$
- traverse the floor copy by induction ($x_n = \varepsilon x_{n-1}$ increases), cross the pair-$n$ edge, traverse the ceiling copy *in reverse* ($x_n = 1 - \varepsilon x_{n-1}$ increases as $x_{n-1}$ decreases)




Start $\mathbf{0}$, end $e_n$. Full verification in the notes.

# From adversarial paths to actual pivot rules

- the path defeats an *adversarially chosen* improving rule; forcing a *specific* rule requires calibrating the instance to the rule — on the instance as stated, Dantzig's rule reaches the optimum in *one* pivot!
- the calibration for Dantzig's rule: the recursive change of variables

  $$x_j = \varepsilon\, x_{j-1} + \theta^{\,j-1}\, \tilde{x}_j \qquad (0 < \theta < \varepsilon),$$

  plus rescaling the ceilings — then Dantzig follows the Hamiltonian path exactly ($\varepsilon = \frac{1}{3}$, $\theta = \frac{1}{9}$: the classical formulation)
- similar constructions for virtually every deterministic rule pre-2000, including steepest edge and shadow vertex (*Goldfarb cube* — exercise)

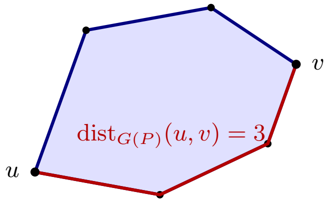
Whether some pivot rule pivots in (strongly) polynomial time: central open question.

# Outline

# A lower bound against every pivot rule

Back to geometric form: $P = \{x : \mathbf{A}x \leq b\}$, $m$ facets; tight set $I(v) := \{i : a_i^\mathsf{T} v = b_i\}$.

- moving between vertices $u, v$ takes $\geq \mathrm{dist}_{G(P)}(u, v)$ pivots — *for every pivot rule*
- combinatorial diameter: $\mathrm{diam}(P) := \max_{u,v \in V(P)} \mathrm{dist}_{G(P)}(u, v)$
- a family with super-polynomial diameter would defeat every pivot rule — none is known




## Polynomial Hirsch conjecture (open)

Is $\mathrm{diam}(P) \leq \mathrm{poly}(m, n)$ for every $n$-dimensional polyhedron $P$ with $m$ facets?

# The Hirsch conjecture, and what is known

- Hirsch, 1957 (letter to Dantzig): is $\mathrm{diam}(P) \leq m - n$?
- for polytopes: stood for over fifty years, until a counterexample [Santos '12]
- all known lower bounds remain within a *constant factor* of $m - n$

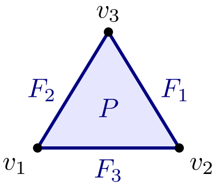
**Theorem (Kalai–Kleitman '92)**

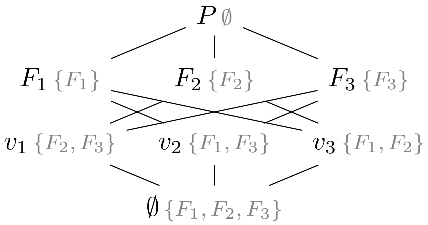
*Every $n$-dimensional polytope $P$ with $m$ facets satisfies* $\mathrm{diam}(P) \leq m^{1+\log_2 n}$.

Goal of this section: prove it — using almost no geometry.

# Vertices as sets of facets: the face lattice

The faces of $P$ ordered by inclusion: the face lattice — all the combinatorics, none of the metric data. Each face $\leftrightarrow$ its set of tight constraints (inclusion *reversing*).




- isomorphic face lattices = *combinatorially equivalent*: e.g. Klee–Minty $P_n \cong [0,1]^n$

# Simple polytopes

Assume $P$ simple: every vertex on exactly $n$ facets. (WLOG for diameter upper bounds: the perturbation $b + (\varepsilon, \ldots, \varepsilon^m)^\mathsf{T}$ from this morning makes $P$ simple, and can only increase the diameter.)

- $v \mapsto I(v)$ is injective into $\binom{[m]}{n}$
- adjacency is set intersection: $uv \in E(P) \iff |I(u) \cap I(v)| = n - 1$
- faces containing $v \leftrightarrow$ subsets of $I(v)$: $J \subseteq I(v)$ gives a face of dimension $n - |J|$

The diameter is determined by the set family $\{I(v) : v \in V(P)\}$ alone — an invitation to discard the geometry.

# Connected layer families

## Definition (CLF)

A *connected layer family of order* $d$ on $N$ symbols: a sequence $L_1, \ldots, L_\ell$ of nonempty collections of $d$-element sets, all sets distinct, such that: *for all* $i \leq j \leq k$ *and* $A \in L_i$, $C \in L_k$, *some* $B \in L_j$ *has* $B \supseteq A \cap C$. The *length* is $\ell$.

| $L_1$ | $L_2$ | $L_3$ | $L_4$ | $L_5$ | $L_6$ | $L_7$ | $L_8$ | $L_9$ | $L_{10}$ |
|---|---|---|---|---|---|---|---|---|---|
| 12 | 13 | 14 | 34 | 35 | 36 | 67 | 68 | 69 | 89 |
| | 24 | 23 | | 46 | 45 | | 79 | 78 | |

Each symbol occupies a consecutive interval of layers: symbol $3$ (red) in layers $2$–$6$, symbol $6$ (green) in $5$–$9$.

# CLF calculus

Two closure properties drive everything (proofs: exercise):

**Lemma (CLF calculus)**

*Let $L_1, \dots, L_\ell$ be a CLF of order $d$ on $N$ symbols.*

(i) *Sublayers: any subsequence $L_{i_1}, \dots, L_{i_r}$ is a CLF of order $d$.*

(ii) *Restriction: for a set $S$ of $\leq d-1$ symbols, the layers containing a set $\supseteq S$ form a contiguous interval; restricting them to those sets and deleting $S$ gives a CLF of order $d - |S|$ on $\leq N - |S|$ symbols, of the same length.*

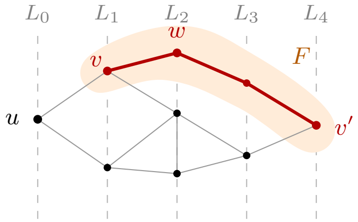
$(ii)$ with $S = \{s\}$: the interval property of the previous slide.

# Polytopes yield CLFs

**Lemma**

*$P$ simple with $m$ facets, $u \in V(P)$: the distance layers $L_i := \{I(v) : \mathrm{dist}_{G(P)}(u,v) = i\}$ form a CLF of order $n$ on $m$ symbols, of length $1 + \max_v \mathrm{dist}(u,v)$.*

- for $A = I(v)$, $C = I(v')$: the face $F$ where $A \cap C$ is tight has a connected graph — the morning's corollary
- a $v$–$v'$ path in $F$ crosses *every* layer in between: at the layer-$j$ crossing, $I(w) \supseteq A \cap C$



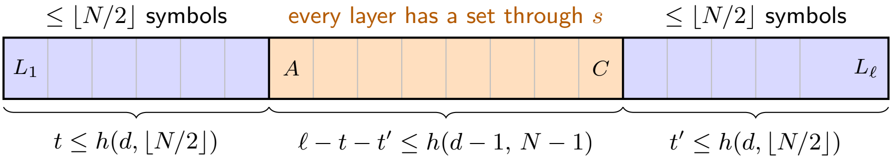

the path in $F$ supplies a witness in every layer

Choosing $u$ in a diameter pair: $\mathrm{diam}(P) \le \max$ CLF length $- 1$.

# The Kalai–Kleitman recursion

$h(d, N) :=$ maximum length of a CLF of order $d$ on $N$ symbols $\quad$ ($0$ if none exists, e.g. $N < d$).

$$h(d, N) \;\le\; 2\,h\big(d, \lfloor N/2 \rfloor\big) \;+\; h(d-1,\, N-1) \qquad (d \ge 2)$$




longest prefix/suffix on $\le \lfloor N/2 \rfloor$ symbols; extended by one layer, each uses $> \lfloor N/2 \rfloor$ symbols $\Rightarrow$ they share a symbol $s$ $\Rightarrow$ restrict & delete $s$: order drops to $d-1$.

# Solving the recursion

Theorem (Abstract Kalai–Kleitman)

$h(d, N) \le N^{1+\log_2 d} = (2d)^{\log_2 N}$ *for* $d \ge 1$, $N \ge 2$. *(Hence* $\mathrm{diam}(P) \le m^{1+\log_2 n}$*.)*

Induction on $N$, all orders at once ($d = 1$: singletons, $h \le N$; $N = 2$: $h \le 2$). Unroll the recursion in its last term until the order reaches $1$:

$$h(d, N) \le 2\sum_{i=2}^{d} h\big(i, \lfloor N/2 \rfloor\big) + N \le 2(d-1)\,(2d)^{\log_2 N - 1} + 2\cdot(2d)^{\log_2 N - 1} = (2d)^{\log_2 N}.$$

- each summand: induction at $\lfloor N/2 \rfloor$ gives $h(i, \lfloor N/2 \rfloor) \le (2i)^{\log_2 (N/2)} \le (2d)^{\log_2 N - 1}$; and $N = 2 \cdot 2^{\log_2 N - 1} \le 2(2d)^{\log_2 N - 1}$
- refined bookkeeping: $\mathrm{diam}(P) \le (m-n)^{\log_2 n}$ [Todd '14] — still quasi-polynomial

# Barnette–Larman: linear in $m$, exponential in $n$

Kalai–Kleitman halves the number of *symbols*; an older argument reduces the *order* instead.

Theorem (Barnette '74, Larman '70)

*Every CLF of order $d$ on $N$ symbols has length $\leq 2^{d-1}N$. Consequently $\operatorname{diam}(P) \leq 2^{n-1}m$.*

- far better in fixed dimension: *linear* in the number of facets
- proof idea: greedily partition the layers into symbol intervals; restriction drops the order to $d-1$ on each; each symbol meets $\leq 2$ consecutive intervals
- guided proof in the exercises

# How long can a connected layer family be?

The abstraction retains almost no geometry — yet no super-polynomial CLF is known.

- longest known: length $\Omega(N^2/\log N)$, with order $d = \Theta(N)$ [EHRR '10]

Hähnle's conjecture (open; posed during Polymath 3)

Every CLF of order $d$ on $N$ symbols has length at most $d(N-1)+1$.

- known in extremal cases; attained for *multiset* layers; even $d = 3$ is open ($4N-1$ vs. conjectured $3N-2$)
- if true: $\mathrm{diam}(P) \leq n(m-1)$ — far stronger than polynomial Hirsch

# Summary and what comes next

**Today:**

- simplex method: correct for every pivot rule (tangent cone); explicit in standard form — edges = kernel directions, slopes = reduced costs
- pivot rules: Dantzig, steepest edge, shadow vertex (trade-off curve!), random facet ($e^{O(\sqrt{n \log m})}$, best known); Klee–Minty forces $2^n - 1$ improving pivots
- the diameter lower-bounds every pivot rule; Hirsch's $m - n$ fails [Santos '12]; polynomial Hirsch wide open
- Kalai–Kleitman: $\mathrm{diam}(P) \leq m^{1+\log_2 n}$, purely combinatorially (CLFs)

**Tomorrow:** why is simplex fast *in practice*? Smoothed analysis: worst-case instances do not survive small random perturbations of the data [Spielman–Teng '04].

*Exercises: removing degeneracy by perturbation, the Goldfarb cube, CLF calculus, Barnette–Larman.*